Petr **Šimčák**

About me

Former theatre and film actor who transitioned into software development through technical education and project-based work. I now focus on practical problem solving in C, C++, and Python, while bringing communication skills, discipline, adaptability, and experience working in demanding team environments.

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Technical skills

- C, C++, Python, Bash, MATLAB
- Git, Docker, Linux, Nginx, LaTeX
- Systems programming, networking, parsing
- Backend development, signal processing

Languages

Czech (C2)
English (C1)
German (B1)

Highlights

2nd place at Škoda Hackathon (team lead / backend role)

Tutor at 42 Prague

3D printer assembly, maintenance, and support

Selected projects

2026–ongoing
FT_TRANSCENDENCE *(full-stack web application)*
◇ Team-based web application developed at 42; current long-term project in a larger collaborative setting.

2026
FT_IRC *(C++98, sockets, poll)*
◇ Built a minimal multi-client IRC server in a team of three; worked mainly on server-side structure and IRC commands.

2025
CUB3D *(C, MLX42)*
◇ Simplified Wolfenstein-style raycasting engine; implemented parsing and core raycasting logic.

2025
BACHELOR THESIS: HEART RATE ESTIMATION FROM PPG SIGNALS *(Python, signal processing)*
◇ Compared multiple methods for heart-rate estimation from physiological signals and worked on automatic signal-quality assessment.

2024–2025
ADDITIONAL 42 PROJECTS *(C, Docker)*
◇ Completed minishell and Inception; gained experience with Unix processes, shell logic, containerization, and deployment setup.

Education

2023–2026
42 PRAGUE *(with period at 42 Heilbronn, 2024–2025)*
◇ Software engineering curriculum based on project work, peer learning, and practical problem solving.

2020–2025
BRNO UNIVERSITY OF TECHNOLOGY
◇ Bachelor's Degree in Sport Technology. Thesis: *Heart Rate Estimation from the PPG Signals*.

Additional experience

2025
ŠKODA B3R BRIGADE — *Team Lead / Backend Developer*
◇ 2nd place at Škoda Hackathon for a talent-matching application built with FastAPI, React, Docker, and SQLite.

2026–present
TUTOR AT 42 PRAGUE
◇ Support other students in understanding projects, debugging issues, and improving problem-solving approaches.

2022–present
3D PRINTER MAINTENANCE AND SUPPORT
◇ Assembled a 3D printer at BUT and later maintained and supported its operation at 42 Prague.

2013–2021
MOVIE ACTOR ( )
◇ Worked in long-term collaborative productions in demanding environments.

2009–2015
THEATRE ACTOR ( )
◇ Professional stage experience in ensemble-based productions.